

Multiple Choice (10 marks)

1. In Python 3, what is the following function returns item from the list with max value?

- (a) `cmp(list)`
- (b) `len(list)`
- (c) `max(list)`
- (d) `min(list)`

2. In Python 3, `b = [11,13,15,17,19,21]; print(b[:2])` outputs what?

- 19,21
- 11,15
- 11,15,19
- 13,17,21

3. In Python 3, which of the following operator in python performs exponential (power) calculation on operands?

- (a) `**`
- (b) `//`
- (c) `is`
- (d) `not in`

4. Which of the following is the hexadecimal representation of the decimal number 231_{10}?

- (a) 17_{16}
- (b) E 4_{16}
- (c) E 7_{16}
- (d) F 4_{16}

5. Which types of functions grow the slowest?

- (a) $O(N^{1/2})$
- (b) $O(N^{1/4})$
- (c) $O(N^{1/N})$
- (d) $O(N)$

True / False (10 marks)

Answer True or False

6. True or False: The addition of the two's-complement numbers 11111111 and 00000001 will result in overflow.
7. True or False: In Python 3, the expression ['Hi!'] * 4 results in a syntax error.
8. True or False: In Python 3, the expression $x \ll 3$ evaluates to 8 when x is equal to 1.
9. True or False: In Python 3, accessing the first element of the tuple ('abcd', 786 , 2.23, 'john', 70.2) will output 'abcd'.
10. True or False: The one-time pad is the only encryption scheme proven to be perfectly secure when used correctly.

Long Form Response (20 marks)

Answer each question with a clear and complete explanation.

11. Write a function to find the most common elements and their counts of a specified text.
12. Write a function to remove words from a given list of strings containing a character or string.

End of Paper